

1 Product Introduction

The handheld RF detection device uses core technologies including spectrum recognition, DJI ID parsing, and Remote ID parsing. It can perform real-time analysis, feature extraction, and target recognition of UAV radio signals across the full 65 MHz-6000 MHz spectrum, enabling accurate UAV target recognition, warning, and trajectory tracking. The device is equipped with a map visualization interface, supports Bluetooth and 4G wireless networking, and can wirelessly transmit alarm data and log information while updating the model feature database online.

The device operates in passive detection mode, only receiving signals without radiation interference. It is lightweight and supports one-handed portable use, making it suitable for police field duty, on-site drone-related case handling, multi-device command coordination, and other mobile security scenarios.



Figure 1 Handheld RF Detection Device

2 Key Features

The device provides full-spectrum UAV detection, multi-model recognition, multi-target tracking and positioning, and other capabilities. Main features are as follows:

- Passive detection uses a signal-receiving-only mode and does not actively transmit electromagnetic signals. It has no radiation, good concealment, and does not interfere with the surrounding electromagnetic environment.
- Full-band coverage supports 65 MHz-6000 MHz custom scanning, covering mainstream UAV control and video transmission communication bands.
- Blacklist/whitelist management allows one-click marking and setting. Compliant UAVs added to the whitelist are automatically shielded and do not trigger alarms, reducing false alarms.
- Intelligent multi-model recognition can identify 800+ UAV models, including mainstream brands such as DJI, Autel, Hubsan, and Xiaomi, and is compatible with frequency-hopping drones, FPV drones, and other non-standard models.
- Synchronous multi-target tracking supports simultaneous detection, positioning, and real-time trajectory tracking of ≥ 10 UAVs, enabling response to multi-target flight scenarios in the same airspace.

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- Rapid airspace photo capture: the device body is equipped with a dedicated photo button to quickly photograph visible UAVs in the airspace with one click and retain on-site evidence.
 - Dual positioning of UAV and pilot supports positioning of mainstream DJI models and RID protocol UAVs, and also provides pilot signal detection and traceability capability.
 - Platform network reporting: the device can access the backend control platform and upload detection alarm data and log records in real time for unified supervision.
 - Automatic spectrum database building: a built-in UAV spectrum feature database supports on-site manual/automatic learning of unknown spectrum signals and autonomous expansion of the model recognition database.
 - PTT network intercom supports wireless networked intercom, enabling instant communication and coordinated dispatch among multiple devices and duty personnel.

3 Basic Specifications

Item	Specification
Operating Principle	RID message reception, protocol parsing, spectrum recognition and detection
Operating Frequency Band	65 MHz-6000 MHz, supports custom scanning
Detection Radius	Urban environment: 1.5-2 km; open environment: 2-3 km
Response Time	≤6 s
Detection Records	>10 records
Startup Time	≤30 s
Screen Size	6 inches
Total Weight	≤650 g
Operating Temperature	-25°C to +40°C